

A long noncoding RNA atlas in Retinitis Pigmentosa

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Introduction: Retinitis Pigmentosa (RP) is a genetically heterogeneous mendelian disorder characterized by rod photoreceptor degeneration and progressive visual impairment and ultimately blindness. The molecular mechanisms underlying photoreceptor death and disease progression are still poorly understood. In that respect, a major gap of knowledge is represented by the role of long noncoding RNAs (lncRNAs), which are known to modulate multiple biological pathways related to human disease pathogenesis. However, there is little information about their role in RP as well as in other Inherited Retinal Disorders (IRDs). Here, we aim at generating a lncRNA expression atlas in the retina across different RP mouse models using next-generation sequencing, and to identify candidates with important roles in retinal homeostasis and diseases.

Methods: Transcriptomic profiling by RNA-seq was performed on the main stages of photoreceptor degeneration (presymptomatic, active rod degeneration and post rod degeneration) in Retinitis Pigmentosa mouse models (rd10, RHO-P347S, RHO-P23H). lncRNAs exhibiting significant expression changes between WT and disease models, and showing consistent deregulation across all three disease models, were selected for further analysis. To predict the potential function of these candidates, lncRNA-mRNA coexpression analysis was carried out. Pairs with the highest correlation score were characterized using Gene Ontology enrichment analysis. To investigate the potential role of the candidate lncRNAs, loss-of-function studies are currently being performed using an *ex vivo* retinal explant model. Subsequently, transcriptomic profiling will assess global gene expression changes following lncRNA knockdown.

Results: Transcriptomic analysis revealed a number of lncRNAs differentially expressed between wild type and RP retina at different disease stages. We focused on the lncRNAs that shared upregulation across all the three models analyzed. Co-expression analysis revealed that some of them are predicted to be associated with functions highly correlated to RP pathogenesis and progression, e.g. inflammation.

Conclusion: We have identified a subset of lncRNAs that display shared dysregulation across different RP models. They represent intriguing candidates to contribute to the pathogenic mechanisms underlying photoreceptor degeneration downstream the primary genetic defect and potential targets for future gene-agnostic therapeutic strategies.

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